Library Management System

EECS 447 — Part 4: Relational Schema Mapping

1. Introduction

Project Overview

The Library Management System stores items, clients, membership types, loans, reservations, notifications, and late fees for a small library. It supports catalog searches, checkouts and returns, borrowing rules, and reports. This part converts the ER model into database tables.

Scope

The schema covers library items, client accounts, membership policies, loans, reservations, notifications, restrictions, and late fees. It defines the tables, attributes, data types, allowed values, keys, and functional dependencies. The project excludes a production interface and connections to outside library systems. SQL implementation and sample data come later.

Glossary

Membership Type — A client category that sets the borrowing limit and daily late-fee rate.

Borrowing Limit — The maximum number of items a client may have checked out at once.

Daily Late-Fee Rate — The amount charged per day for an overdue item.

Domain — The values allowed in a field, such as permitted account statuses.

Functional Dependency (FD) — A rule where one value determines another, such as Membership Type ID determining the borrowing limit.

Referential Integrity — The rule that linked records must exist, such as a loan referring to an existing client.

2. Relational Schema Mapping

ER to Relational Mapping Approach

Each entity becomes a table, with its attributes as columns and its ID as the primary key. Library Item stores books, digital media, and magazines, identified by Item Type.

For a one-to-many relationship, the table on the many side stores the other table's primary key as a foreign key. Late Fee stores a required, unique Loan ID, allowing at most one fee per loan.

Loan and Reservation link clients to items. Each has its own ID so the same client can borrow or reserve an item more than once.

Relations and Primary Keys

PK = primary key; FK = foreign key.

Relation	Attributes
Membership Type	Membership Type ID (PK), Name, Borrowing Limit, Daily Late-Fee Rate
Client	Client ID (PK), First Name, Last Name, Phone Number, Membership Type ID (FK), Account Status
Library Item	Item ID (PK), Title, Item Type, Availability Status, Borrowing Restricted, Author/Creator, ISBN, Publication Year, Genre, Issue Number, Publication Date
Loan	Loan ID (PK), Item ID (FK), Client ID (FK), Checkout Timestamp, Due Date, Return Timestamp
Late Fee	Fee ID (PK), Loan ID (FK, UNIQUE), Amount Owed, Amount Paid
Reservation	Reservation ID (PK), Client ID (FK), Item ID (FK), Reservation Date, Reservation Status
Notification	Notification ID (PK), Client ID (FK), Loan ID (FK, nullable), Reservation ID (FK, nullable), Notification Type, Message, Creation Timestamp

Primary keys are unique and required. Late Fee.Loan ID is also unique and required. Item ID identifies each library item; ISBN, names, and phone numbers are not unique identifiers.

Attributes and Domains

The data dictionary lists each attribute's type, allowed values, and constraints. Required text cannot be blank.

Foreign Keys and Relationship Constraints

Foreign key	Referenced primary key	Participation
Client.Membership Type ID	Membership Type.Membership Type ID	Required; one membership type per client.
Loan.Item ID	Library Item.Item ID	Required; one item per loan.
Loan.Client ID	Client.Client ID	Required; one client per loan.
Late Fee.Loan ID	Loan.Loan ID	Required and UNIQUE; zero or one fee per loan.
Reservation.Client ID	Client.Client ID	Required; one client per reservation.
Reservation.Item ID	Library Item.Item ID	Required; one item per reservation.
Notification.Client ID	Client.Client ID	Required; one recipient per notification.
Notification.Loan ID	Loan.Loan ID	Required for loan notices; otherwise null.
Notification.Reservation ID	Reservation.Reservation ID	Required for reservation notices; otherwise null.

Every non-null foreign key must refer to an existing record. Records cannot be deleted while other records reference them. A loan or reservation may have multiple notifications.

Functional Dependencies

The arrow (→) means that the value on the left determines the values on the right.

Relation	Functional dependencies
Membership Type	Membership Type ID → Name, Borrowing Limit, Daily Late-Fee Rate
Client	Client ID → First Name, Last Name, Phone Number, Membership Type ID, Account Status
Library Item	Item ID → Title, Item Type, Availability Status, Borrowing Restricted, Author/Creator, ISBN, Publication Year, Genre, Issue Number, Publication Date
Loan	Loan ID → Item ID, Client ID, Checkout Timestamp, Due Date, Return Timestamp
Late Fee	Fee ID → Loan ID, Amount Owed, Amount Paid; Loan ID → Fee ID, Amount Owed, Amount Paid
Reservation	Reservation ID → Client ID, Item ID, Reservation Date, Reservation Status
Notification	Notification ID → Client ID, Loan ID, Reservation ID, Notification Type, Message, Creation Timestamp

Membership Type ID determines the borrowing limit and daily fee rate. For notifications linked to loans, Loan ID determines Client ID; for reservation notices, Reservation ID determines Client ID.

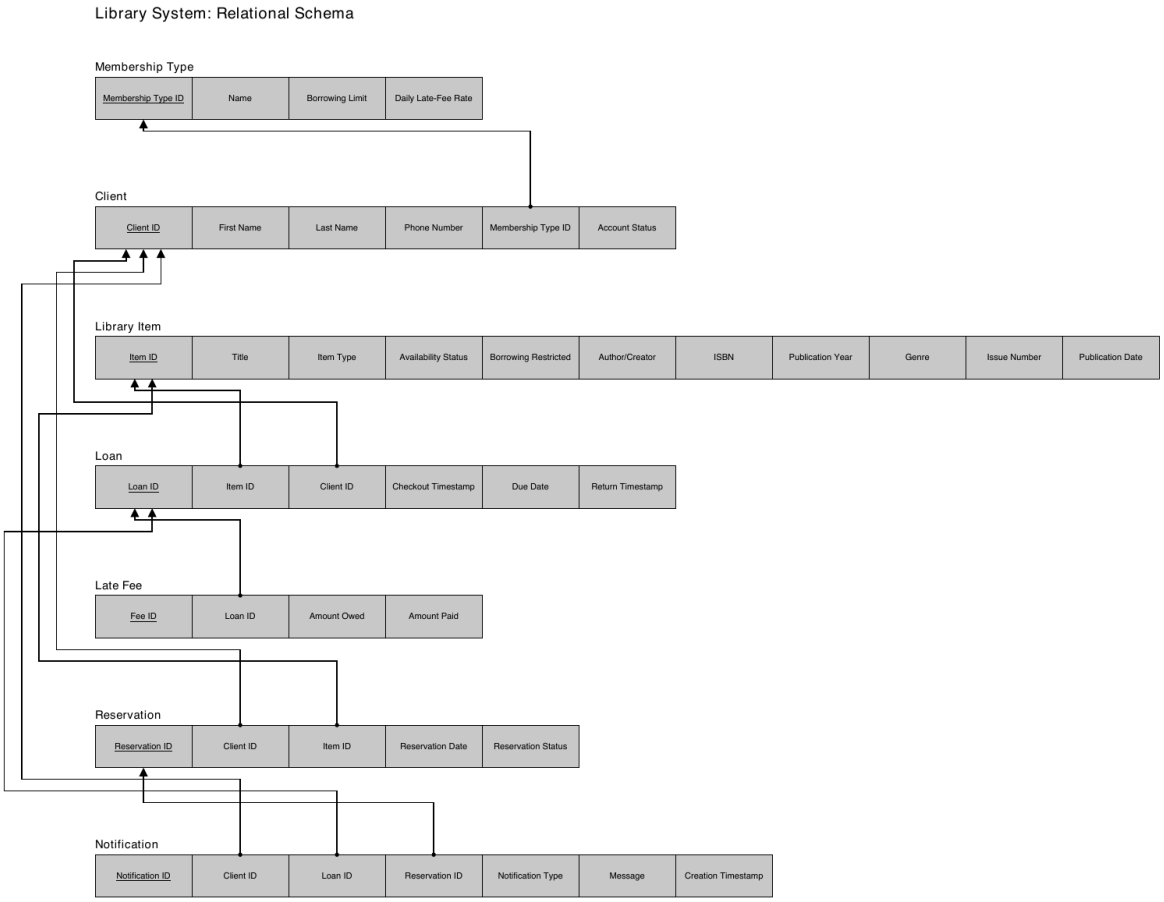
Data Integrity and Business Rules

- An item can have only one active loan at a time. A null Return Timestamp identifies an active loan.
- Checkout must respect the membership borrowing limit, client account restrictions, and item borrowing restrictions.
- Due Date must be on or after the checkout calendar date. Return Timestamp, when present, must be on or after Checkout Timestamp.
- Reservations are for unavailable items. Each reservation identifies one existing client and item and uses a permitted status.
- Each notification references exactly one loan or reservation, as determined by its type, and its Client ID must match the client on that record.

- A loan has at most one late-fee record. Amount Owed and Amount Paid cannot be negative. The remaining balance is Amount Owed minus Amount Paid.
- Availability Status must stay consistent with loan and reservation records.

3. Relational Schema Diagram

Underlined attributes are primary keys. Arrows connect foreign keys to the primary keys they reference.



4. Schema Documentation & Data Dictionary

PK = primary key; FK = foreign key; NN = required. IDs are integers from 1 to 2,147,483,647. Required text cannot be blank. Nullable fields may be left empty where stated.

Membership Type

Attribute	Data type	Domain and constraints	Description
Membership Type ID	INTEGER	PK; NN; positive ID	Identifies a borrowing-policy category.
Name	VARCHAR(50)	NN; nonblank	Membership label, such as regular, student, or senior; not assumed unique.
Borrowing Limit	INTEGER	NN; 0–2,147,483,647	Maximum simultaneous active loans.
Daily Late-Fee Rate	NUMERIC(6,2)	NN; 0.00–9,999.99	Daily monetary charge for an overdue loan.

Client

Attribute	Data type	Domain and constraints	Description
Client ID	INTEGER	PK; NN; positive ID	Identifies one library client.
First Name	VARCHAR(100)	NN; nonblank	Client’s given name.
Last Name	VARCHAR(100)	NN; nonblank	Client’s family name.
Phone Number	VARCHAR(32)	NN; nonblank	Contact number; permits international prefixes, spaces, punctuation, and extensions.
Membership Type ID	INTEGER	NN; FK → Membership Type.Membership Type ID	Client’s current membership category.

Account Status	VARCHAR(10)	NN; active, inactive, restricted	Controls account eligibility.

Library Item

Attribute	Data type	Domain and constraints	Description
Item ID	INTEGER	PK; NN; positive ID	Identifies an individual collection item.
Title	VARCHAR(255)	NN; nonblank	Displayed catalog title.
Item Type	VARCHAR(13)	NN; book, digital_media, magazine	Selects the applicable descriptive fields.
Availability Status	VARCHAR(12)	NN; available, on_loan, reserved	Current circulation state; synchronized with loans and reservations.
Borrowing Restricted	BOOLEAN	NN; TRUE or FALSE	Indicates an item-specific prohibition on checkout.
Author/Creator	VARCHAR(255)	Required, nonblank for books/digital media; null for magazines	Creator credit for the item.
ISBN	VARCHAR(13)	Nullable; if present, 10 or 13 characters in normalized ISBN form	Optional edition identifier; not UNIQUE. Store without spaces or hyphens.
Publication Year	SMALLINT	Required for books/digital media; 1–9999; null for magazines	Year of publication.

Genre	VARCHAR(100)	Required, nonblank for books/digital media; null for magazines	Catalog genre label.
Issue Number	VARCHAR(50)	Required, nonblank for magazines; null otherwise	Magazine issue designation; may include letters or punctuation.
Publication Date	DATE	Required for magazines; valid calendar date; null otherwise	Calendar publication date of the issue.

Loan

Attribute	Data type	Domain and constraints	Description
Loan ID	INTEGER	PK; NN; positive ID	Identifies one checkout event.
Item ID	INTEGER	NN; FK → Library Item.Item ID	Borrowed item; unique among active loans.
Client ID	INTEGER	NN; FK → Client.Client ID	Client responsible for the loan.
Checkout Timestamp	TIMESTAMP WITH TIME ZONE	NN; valid timestamp	Actual checkout instant.
Due Date	DATE	NN; valid calendar date; on/after checkout date	Last calendar day before the loan becomes overdue.
Return Timestamp	TIMESTAMP WITH TIME ZONE	Nullable; valid timestamp; on/after checkout	Null while active; records the return instant.

Late Fee

Attribute	Data type	Domain and constraints	Description
Fee ID	INTEGER	PK; NN; positive ID	Identifies one fee assessment.
Loan ID	INTEGER	NN; UNIQUE; FK → Loan.Loan ID	Overdue loan associated with the fee.
Amount Owed	NUMERIC(10,2)	NN; 0.00–99,999,999.99	Total assessed charge before payments, not the remaining balance.
Amount Paid	NUMERIC(10,2)	NN; 0.00–99,999,999.99	Payments applied; balance = Amount Owed − Amount Paid.

Reservation

Attribute	Data type	Domain and constraints	Description
Reservation ID	INTEGER	PK; NN; positive ID	Identifies one reservation event.
Client ID	INTEGER	NN; FK → Client.Client ID	Client making the request.
Item ID	INTEGER	NN; FK → Library Item.Item ID	Requested individual item.
Reservation Date	DATE	NN; valid calendar date	Request date in the library time zone.
Reservation Status	VARCHAR(10)	NN; pending, available, fulfilled, cancelled	Current request stage.

Notification

Attribute	Data type	Domain and constraints	Description
Notification ID	INTEGER	PK; NN; positive ID	Identifies one notice.
Client ID	INTEGER	NN; FK → Client.Client ID	Recipient; must match the linked event's client.
Loan ID	INTEGER	Nullable; FK → Loan.Loan ID	Required for upcoming-due or overdue notices; null for reservation notices.
Reservation ID	INTEGER	Nullable; FK → Reservation.Reservation ID	Required for reservation-availability notices; null for loan notices.
Notification Type	VARCHAR(21)	NN; upcoming_due, overdue, reservation_available	Determines which event reference is required.
Message	TEXT	NN; nonblank	Notification message.
Creation Timestamp	TIMESTAMP WITH TIME ZONE	NN; valid timestamp	Time the notice record was created.

Books and digital media require Author/Creator, Publication Year, and Genre; ISBN is optional. Magazines require Issue Number and Publication Date. Non-applicable fields are null.

Upcoming-due and overdue notifications require Loan ID and a null Reservation ID. Reservation-availability notifications require Reservation ID and a null Loan ID.

Project Meeting Log

Date: October 18th

Time: 5-6pm

Location: Virtual

Objective: Delegate tasks for Part 4 of the project, focusing on expanding the ER diagram into a relational schema, defining relationships and constraints, and organizing the required documentation and deliverables.

Team Members Present: Mo, Pranav, Gabe, Karim, Daniel

Tasks Completed

- Gabe: Created the relational schema diagram
- Pranav: Completed project documentation and reviewed the final deliverables
- Mo: Reviewed documentation and schema diagram
- Daniel: Converted ER diagram entities into relational tables
- Karim: Mapped relationships and identified foreign keys

Tasks Allocated

- Gabe: Create relational schema diagram
- Pranav: Complete project documentation and review final submission
- Mo: record notes and assist with documentation
- Daniel: Convert ER diagram into relational tables
- Karim: Map relationships and identify foreign keys

Deadline: Sunday 10pm

Follow-Up Action

Review and Submit by Sunday 11:59